NutriVision

AI-POWERED FOOD ANALYSIS

AI-Powered Indian Food Nutrition Estimator

A complete end-to-end AI system that photographs a meal and returns a full nutritional breakdown — designed specifically for Indian multi-dish cuisine.

97.92%	79	79	6-Stage
Top-1 Accuracy	Food Classes	Supported Dishes	AI Pipeline

IIT Madras · Data Science & AI Lab

Group 2 · Milestone 6 Final Report

Why Existing Nutrition Trackers Fall Short

Manual Entry

Users must type every food item by hand — tedious, error-prone, and rarely done consistently.

Fixed Portions

All existing tools assume a fixed 100g serving, ignoring real-world portion sizes.

Poor Indian Coverage

Most apps treat 'biryani' as one item, with no support for thali plates or regional varieties.

No Personalisation

Recommendations are generic — not tailored to an individual's dietary deficiencies.

NutriVision solves all four — using a single photo + AI pipeline designed for Indian cuisine

Project Objectives

1. Detect & classify multiple food items in a single image
2. Estimate real-world portion weight from a 2D photo
3. Map foods to a structured Indian nutrition database
4. Perform personalised nutrient gap analysis
5. Generate dietary suggestions via LLM pipeline
6. Deploy as a live, end-to-end web application

NutriVision vs. Alternatives

Feature	Traditional	Basic AI	NutriVision
Manual entry	Yes	Yes	None
Multi-food image	No	Limited	Full
Portion estimation	No	Fixed 100g	PCA+Density
Indian cuisine	Partial	Partial	79 classes
Personalised RDA	Limited	No	Nutrition.json

USER GUIDE – APPLICATION OVERVIEW

Key features, requirements, and access instructions

Access the Application

- Browser-based web app (no login required)
- Open directly in Chrome / Edge / Firefox
- Upload a food image and analyse in one workflow

What the User Needs

- Stable internet connection
- Clear food image (JPG / JPEG / PNG)
- ₹10 coin placed near the food for scale calibration
- Modern browser for smooth performance

Main Interface Includes

- Upload panel (upload photo or use camera)
- Analyse button
- Detected food items panel
- Pipeline status checklist
- Nutrition summary table
- LLM weight reasoning section

Single photo + ₹10 coin → food detection + weight estimation + nutrition breakdown

USER GUIDE – STEP-BY-STEP USAGE

Simple 5-step workflow for end users

Step 1 – Open the Web App

- Launch the deployed NutriVision application in a browser

Step 2 – Prepare the Food Image

- Use a clear, well-lit image
- Prefer top-down / near top-down angle
- Keep background simple and uncluttered

Step 3 – Place a ₹10 Coin

- Place the ₹10 coin near the food
- Keep it fully visible and flat on the same surface
- Used for real-world size calibration (2.7 cm reference)

Step 4 – Upload or Use Camera

- Click “Upload photo or use camera”
- Upload an image from device or capture a new image
- Verify that both food + coin are clearly visible

Step 5 – Click Analyse

- Starts the complete AI pipeline automatically
- No manual intervention needed

USER GUIDE – AI PIPELINE & OUTPUT INTERPRETATION

What happens after the user clicks Analyse

NutriVision Processing Pipeline

- SAM3 Segmentation – isolates food region
- Coin Scale Detection – detects ₹10 coin for scale
- Food Classification – predicts food category
- PCA Geometry Estimation – extracts real-world dimensions
- Weight & Nutrition Estimation – calculates grams + nutrients
- LLM Weight Reasoning – explains how the estimate was derived

What the User Sees

- Segmented food image with predicted label
- Estimated weight displayed on image
- Number of detected food items
- Approximate total calories

Output Panels

- Detected Food Items
- Nutrition Summary Table
- LLM Weight Reasoning

USER GUIDE – UNDERSTANDING THE FINAL RESULTS

How to interpret the generated output

1. Detection Summary

- Number of detected food items
- Approximate total calories

2. Nutrition Summary Table

- Food name
- Weight (g)
- Calories (kcal)
- Carbohydrates / Protein / Fat
- Free Sugar / Fibre
- Sodium / Calcium / Iron

3. LLM Weight Reasoning Panel

- Predicted food name
- Confidence score
- Estimated weight
- Assumed geometric shape
- Density used
- Reasoning behind the estimate

Key Benefit: Not just prediction — the app explains why it predicted that weight

USER GUIDE – BEST PRACTICES, FAQs & TROUBLESHOOTING

Practical tips for better accuracy and common issue resolution

Best Practices

- Use good lighting
- Prefer top-down angle
- Keep ₹10 coin fully visible
- Place coin close to the food
- Avoid blurry images
- Keep food items separated if multiple dishes
- Use a simple background

FAQs

- Is ₹10 coin mandatory? Strongly recommended
- Supported formats? JPG / JPEG / PNG
- Login required? No
- Can multiple items be analysed? Yes
- Clearer separation gives better results

Troubleshooting

- Image upload failed → Check format / refresh browser
- Analyse button not working → Ensure image preview is visible
- Coin not detected → Keep coin fully visible and flat
- Wrong food prediction → Retake with better lighting & angle
- Nutrition table missing → Wait for all pipeline stages / retry
- Unrealistic weight → Ensure correct coin placement and top-down image

Title and Idea

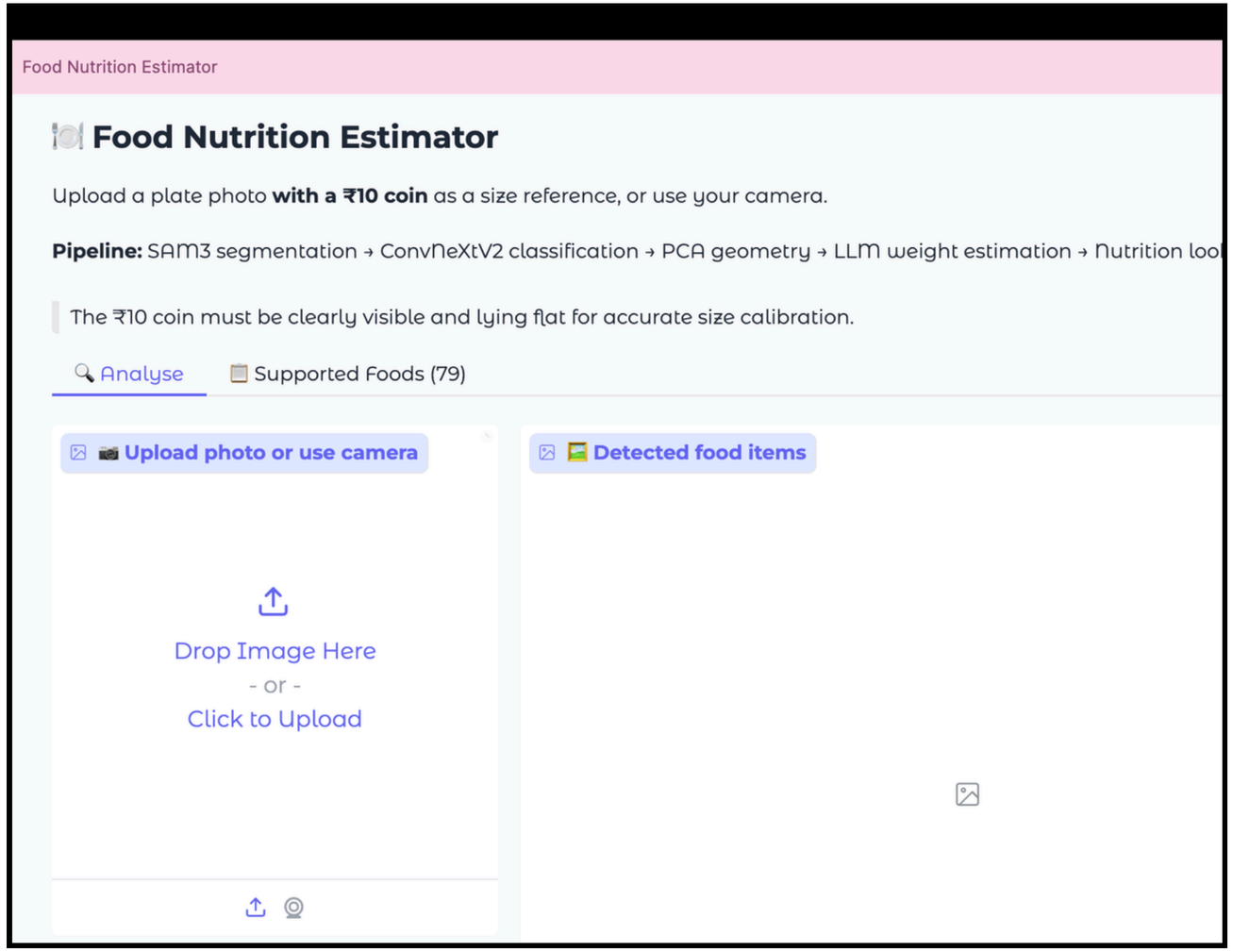
NutriVision: AI-Based Food Nutrition Estimator

- An easy-to-use application for analyzing food from images
- Just upload a food photo → get nutrition instantly
- No manual calorie tracking required

What it does:

- Takes a food image
- Identifies the food
- Estimates calories & nutrients

Makes nutrition tracking simple and accessible



Purpose & Use Cases

Purpose

- Help users make healthier food choices
- Simplify nutrition tracking
- Show real-world use of AI in daily life

Who can use it?

- Students learning nutrition
- Fitness enthusiasts tracking calories
- Individuals managing diet
- Anyone curious about food nutrition

How It Works (Simple View)

1. Upload or take a photo of food
2. Place a ₹10 coin for size reference
3. Click Analyze
4. System processes the image
5. Results are displayed instantly

Input:

- Food image (clear, well-lit)

Output:

- Food name
- Weight (grams)
- Calories
- Nutrients (protein, fats, etc.)

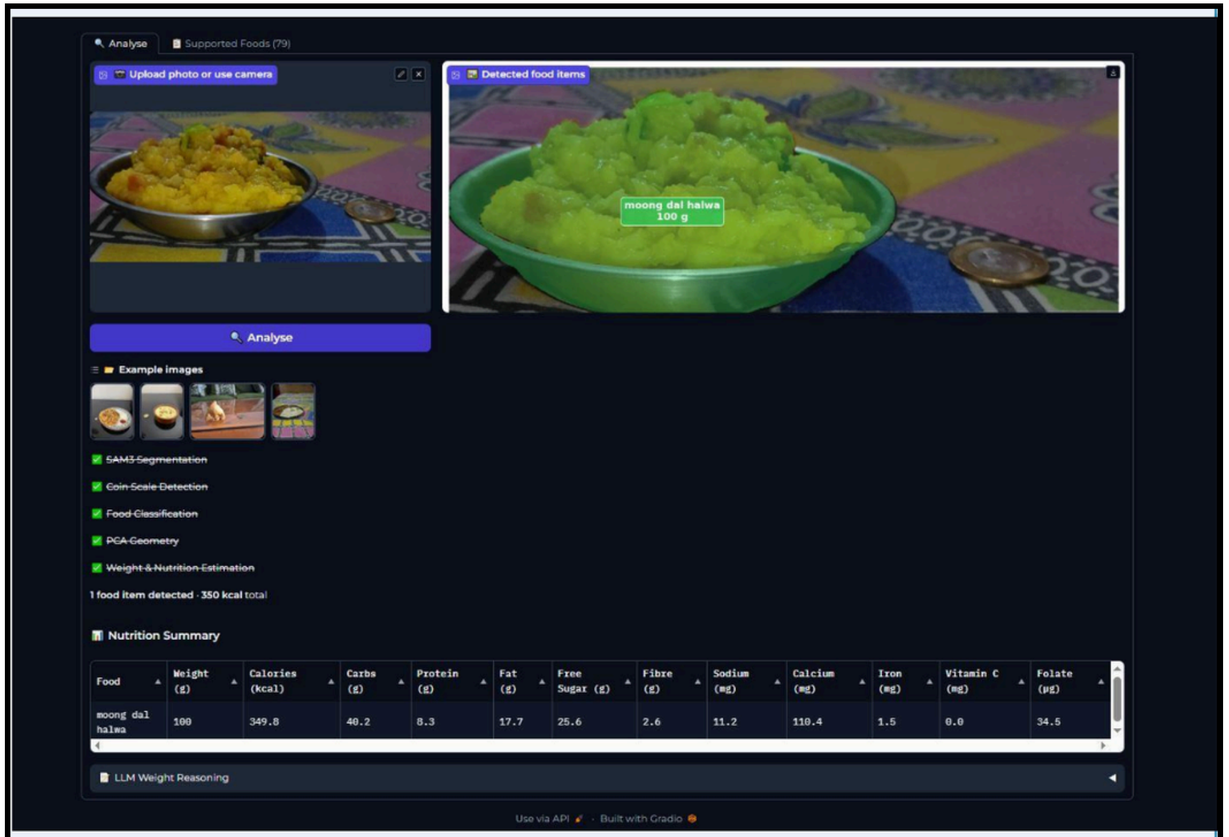


Figure 2.1c *App UI — successful detection of moong dal halwa (100 g, 350 kcal) with full nutrition table*

Features, Benefits and Limitations

Key Features

1. Supports ~79 Indian food dishes
2. Simple web interface (no technical knowledge needed)
3. Works with single or multiple food items

Benefits

1. Saves time (no manual tracking)
2. Easy and user-friendly
3. Encourages healthy eating habits

Impact

1. Makes nutrition awareness accessible to everyone
2. Bridges gap between AI and daily life

Troubleshooting/Limitations

If the image is not detected properly:

- Try using a clearer image
- Avoid multiple food items in one image
- Place a ₹10 coin flat and clearly visible in frame
- Use good, even lighting
- Avoid extreme angles (overhead or top-down works best)

Training Data — Multi-Track Approach

KHANA DATASET (Classification)

Images:	131,819
Classes:	80 Indian foods
Split:	80/10/10 stratified
Avg per class:	~164.8 images
Imbalance:	Min 12 — Max 788

DETECTION DATASETS (YOLO, M1–M3)

IndianFood10 (Roboflow)	**11,547 imgs**
IndianFoodNet-30	**5,446 imgs**
South Indian Food Det.	**9,338 imgs**
Indian Food YOLOv5	**~14,000 imgs**
TOTAL (merged)	**39,185 · 118 classes**

Nutrition DB

310+ Indian dishes
per-100g macros & micros

Custom Weight Data

40 overhead plate photos
with kitchen scale labels and with 10 rs coin

From Detection-First → Segmentation-First Architecture

OLD PIPELINE (Milestones 1–3)

1. **YOLO Detection**
   Bounding boxes for food items
2. **SAM-2 Segmentation**
   Prompted by YOLO boxes
3. **MLP Weight Est.**
   Pixel ratio → grams regression
4. **CSV + Pinecone RAG**
   Nutrition lookup
5. **Gemini API**
   Generic dietary suggestions

REDESIGNED →

NEW PIPELINE (Milestones 4–6)

1. **SAM3 Segmentation**
   Direct pixel masks — no YOLO needed
2. **Coin Scale Detection**
   ₹10 coin = 2.7cm calibration reference
3. **ConvNeXtV2-Tiny**
   97.92% accuracy — 80 Indian food classes
4. **PCA Geometry**
   Rotation-invariant width/height extraction
5. **LLM Volume + Density**
   Ollama-Qwen + hard JSON density enforcement
6. **Nutrition Scaling**
   Per-100g → actual weight, Nutrition.json

Classification Backbone Comparison

ConvNeXtV2-Tiny (CHOSEN)

Parameters:	27.9M
Input Size:	176 × 176 px
Optimiser:	AdamW (head 1e-3 / backbone 1e-4)
Regularisation:	Dropout 0.2 + early stopping
Augmentation:	RandomResizedCrop + Albumentations
Loss:	CrossEntropy + label smoothing ε=0.1
LR Schedule:	1-epoch warmup → cosine decay
Throughput:	150.7 images/sec (Tesla T4)

EfficientNetV2-S (Comparison)

Parameters:	20.3M
Input Size:	300 × 300 px
Phase 1:	3 epochs — frozen backbone
Phase 2:	7 epochs — full fine-tune
Augmentation:	MixUp + CutMix (Phase 2)
Normalisation:	mean/std = [0.5, 0.5, 0.5]
Loss:	CrossEntropy
Throughput:	109.2 images/sec (Tesla T4)

Classification Performance — Hold-out Test Set (13,182 images)

97.92%
ConvNeXtV2
Top-1 Accuracy

99.76%
ConvNeXtV2
Top-5 Accuracy

0.9746
ConvNeXtV2
Macro F1

0.9859
ConvNeXtV2
mAP Score

Model	Top-1	Top-5	Precision	Recall	Macro F1	mAP
EfficientNetV2-S	92.85%	99.12%	0.9251	0.8877	0.9024	0.9387
ConvNeXtV2-Tiny ★	**97.92%**	**99.76%**	**0.9746**	**0.9753**	**0.9746**	**0.9859**

Micro-AUC: EfficientNetV2-S = 0.999 · ConvNeXtV2-Tiny = 1.000

Where Models Fail — and Why

Top Confusion Pairs (both models)

Confused Pair	EffNet-S	ConvNeXt
Paneer pizza → Chicken pizza	48	24
Chicken pizza → Paneer pizza	38	11
Chicken pizza → Margherita pizza	20	9
Phirni ↔ Kheer	Some	Negligible

Root Cause: Pizza variants share near-identical 2D visual features — only ingredient-level labels or multi-view images would resolve this.

Strongest Classes (ConvNeXt)

- Steamed momo — F1 = 0.994
- Sabudana vada — F1 = 0.989
- Grilled sandwich — F1 = 0.983

Weakest Classes (EffNet-S)

- Amritsari kulcha — F1 = 0.486
- Phirni — F1 = 0.645
- Moong dal halwa — F1 = 0.686

Top-5 accuracy = 99.53–99.76% confirms the correct class is almost always in the model's top 5 predictions — the learned representations are semantically well-organised.

Prerequisites

Lightning AI environment

Used to run and deploy the ML application with GPU support

Hugging Face Token

Used to access and download pre-trained machine learning models (SAM-3)

Ollama API Key

Used to connect the application with large language models for AI-powered responses.

Clone the repository

Lightweight Clone Repo

```
git clone --filter=blob:none --sparse
https://github.com/23f3001764/Group-2-DS-and-AI-Lab
Project.git

cd Group-2-DS-and-AI-Lab-Project

git sparse-checkout set nutrivision_codes

cd nutrivision_codes
```

Environment Configuration

Authentication key for Hugging Face

```
HF_TOKEN=<your_huggingface_token>
```

Access key for Ollama LLM services

```
OLLAMA_API_KEY=<your_ollama_api_key>
```

Installation and Model setup

Install all required dependencies:

```
pip install -r requirements.txt

cd files_models
gdown https://drive.google.com/file/d/1w8PQt-
7Ofn4pjTUUC2YqwIxEicy2Zxhe/view?usp=sharing
cd ..
```

Running the app

- Activate the GPU T4
- Execute the application using:
  - python app.py
- First, download & Install the Gradio
- Open the Gradio and set the port to 7860, and in the running command, python app.py, and choose the T4 GPU
- Configure the deployment with:
  - Framework: Gradio
  - Mode: Serverless
  - Port: 7860
- Once deployed, the application will be accessible via a public endpoint.

Module Description

- app.py: Main application entry point Builds and launches the Gradio interface Handles user input and displays output
- pipeline.py: Core logic for inference Handles: Image preprocessing Model prediction Post-processing
- models.py: Loads trained model and weights. Ensures the model is ready for inference
- config.py: stores File paths, Constants, Environment configurations
- food_nutrition.json → Maps food items to nutritional values
- food_density.json → Supports portion/density-based calculations

Key Technical Hurdles Encountered & Resolved

C1

CUDA Out-of-Memory on phone photos

Root cause: 3000–4000px phone photos exceed SAM3 native resolution with 4× queries

Auto-resize to 1008px before SAM3 inference using LANCZOS resampling

C2

LLM chose arbitrary density values

Root cause: Density not in the prompt — LLM guessed from training knowledge

Hard-enforced food_density.json — LLM output is overwritten post-parse

C3

LLM parse failures crashed the whole pipeline

Root cause: Unstructured LLM response fails Pydantic schema validation

Graceful fallback + error note; pipeline continues for other items

C4

Double-counting food items in thali plates

Root cause: SAM3 run with 3 prompts; all returning overlapping masks

IoU clustering + containment threshold — one mask per distinct food region

From MLP Regression → PCA + Geometry + LLM

Approaches Considered & Rejected

ZoeDepth / MiDaS

Liquid dishes (dal, sambar) have unreliable depth maps

Volume Reconstruction

Flat foods (roti, dosa) give near-zero height

MLP Regression

Camera-distance-dependent; needs large labelled set

Why PCA Wins

- No depth sensors or special hardware needed
- Works for both flat breads AND 3D items
- Interpretable — humans can verify the geometry
- Decouples volume from density estimation

CHOSEN: PCA + Coin Calibration

Coin Calibration

₹10 coin (2.7cm) → pixels-per-cm scale factor

PCA on Mask Pixels

Major/minor axes = rotation-invariant width & height

Real-World Dimensions

Pixel measurements × px-per-cm scale factor

Geometric Volume

Cylinder (round) or cuboid (flat bread) approximation

Density Enforcement

food_density.json → overrides LLM guess (hard-coded)

Weight Calculation

Weight (g) = Volume (cm^3) × Density (g/cm^3)

Live Gradio Web App — NutriVision

Live URL: https://7860-01kp8gvr84ng3myeemyw7vzz46.cloudspaces.litng.ai

5-Step Streaming Pipeline

- Step 1 — SAM3 Segmentation
- Step 2 — Coin Scale Detection
- Step 3 — Food Classification
- Step 4 — PCA Geometry
- Step 5 — Weight & Nutrition

Sample Detection Results

	Detected As	Weight	Calories	Status
	Moong dal halwa	100g	350 kcal	Misclassified
	Steamed momo	37g	52 kcal	Correct
	Aloo Gobi (thali)	193g	205 kcal	Correct
	Biryani → Chivda	25g	28 kcal	Misclassified
	Roti → Garlic Naan	19g	23 kcal	Visual overlap

Stack: Python · PyTorch · timm · HuggingFace SAM3 · Gradio · LangChain · Ollama-Qwen · Lightning AI GPU

What We Learned

Key Findings

- ConvNeXtV2-Tiny is the strongest classifier — 97.92% Top-1, outperforming EfficientNetV2-S by ~5%
- Effective batch size (via gradient accumulation) was the most impactful hyperparameter: +3.44 Val F1
- Conservative backbone LR is critical — reducing from 1e-4 to 5e-5 gave +3.30 F1 points
- PCA + coin calibration is more practical than depth estimation or MLP for Indian food weight
- Preprocessing bugs caused the biggest misleading result — wrong size/normalisation = apparent random model
- Top-5 accuracy of 99.76% shows semantically well-organised representations

Limitations

- Pizza-variant confusion (chicken/paneer/margherita) — visually irreducible in 2D
- Minority classes with <30 test samples show high metric variance
- Dataset may not cover all real-world conditions (dim light, partial occlusion)
- PCA weight estimate depends on clean segmentation and correct shape assumptions
- Macro ROC-AUC numerical issues with 80 imbalanced classes

Roadmap for Further Development

Classification

- use user input for class of food
- Expand to 150+ classes incl. regional specialties

Weight Estimation

- Multi-view input for tall glasses (lassi, chaas)
- fine tuned model for better results

App & Deployment

- use iphone arkit to estimate pixels per cm
- Multi-meal daily tracking vs ICMR RDA totals
- User profile onboarding (age, gender, activity level)
- Model quantisation (INT8/FP16) for smaller GPU inference

Data & Nutrition

- Cooking-method-aware nutritional values (fried vs steamed)
- Regional variation — Chettinad vs Kolkata biryani macros differ
- I

Web App Presentation

Link 1: click here
Link 2 with new update: click here

NutriVision

GROUP 2 · IIT MADRAS

Team Members

- Sahil Raj
- Sahil Sharma
- Aman Mani Tiwari
- Samreen Fathima
- Tasneem Shahnawaz

GitHub

github.com/23f3001764/Group-2-DS-and-AI-Lab-Project

Key Achievements

97.92%	Top-1 accuracy across 80 Indian food classes — ConvNeXtV2-Tiny
Novel	Coin-calibrated PCA geometry for depth-sensor-free portion estimation
6-Stage	End-to-end pipeline from photo to personalised nutritional report
Live	Production Gradio app on Lightning AI with GPU acceleration & streaming

NutriVision — Smarter nutrition tracking, designed for India